

Software Engineering Lab 1

Lab 1 – Requirements Engineering & UML Use-Case Modelling

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Use-Case Flow Specification

Problem Statement #56: Peer Skill Exchange & Mentorship Network

Use Case: UC-04 – Book Skill Exchange Session

Actors

Primary: Learner Member. Secondary: Skill Mentor, External Calendar System.

Preconditions

1. The Learner Member is registered and logged in.
2. The Learner Member has a time-credit balance of at least 1 hour.
3. The Skill Mentor has published at least one available time slot.

Postconditions

1. On success: the session slot is reserved on both members' calendars, the learner's time credit is placed on hold for the session duration, and both parties receive a confirmation.
2. On failure: no slot is reserved and no time credit is held.

Main Success Scenario

1. Learner Member searches for and selects a Skill Mentor's listing.
2. System displays the Mentor's published available time slots.
3. Learner Member selects a desired time slot and session duration.
4. System checks that the Learner Member's time-credit balance is sufficient for the session duration.
5. Learner Member confirms the booking request.
6. System reserves the slot and notifies the Skill Mentor of the pending request.
7. Skill Mentor accepts the booking request.
8. System places a hold on the Learner Member's time-credit balance equal to the session duration.
9. System optionally syncs the confirmed session to the Learner's and Mentor's external calendars (extends UC-05: Sync Session with External Calendar).
10. System sends a confirmation notification to both the Learner Member and the Skill Mentor.
Use case ends successfully.

Alternate Flow A – Insufficient Time-Credit Balance (branches at step 4)

1. System detects that the Learner Member's balance is less than the requested session duration.
2. Booking is blocked and the Learner Member is shown an "Insufficient credits" message with a link to view options for earning credits.
3. Use case ends without a reservation being made.

Alternate Flow B – Mentor Declines the Request (branches at step 7)

1. Skill Mentor declines the booking request.
2. System releases the reserved slot back to the Mentor's available slots.

3. Learner Member is notified that the request was declined.
4. Learner Member is prompted to select a different time slot or a different mentor.